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Vikas Kumar

Data Scientist

GitHub: vikasmech
LinkedIn: vikasmechnitk

SKILLS

Tools and Languages Python, Git, R, SQL, BigQuery, Elasticsearch, Matlab, GitLab CI/CD, Jira
Library Numpy, Pandas, ScikitLearn, Keras, Pytorch, TensorFlow, spaCy, pySpark
Other Machine Learning, Deep Learning, Statistical Analysis, Research

WORK EXPERIENCE

Senior Data Scientist

Jan 2023 — Present

DS team, Pocketfm

Bangalore

- Module Rank/Position Personalization with Dedup, enhancing feed personalization for the platform.
 - Successfully designed and executed strategies to optimize content recommendations, leading to a 4% increase in user engagement (playtime Hr).
- Show wise Cohort based Average revenue per listener (ARPL) prediction for feed show ranking
 - Used Auto regressive data structure to predict ARPL@hN+10 based on ARPL@hN and achieved MAPE of 2% across show language, locale, gender.
 - Projected ARPL @ 100 hour based show ranking was used to create feed recommendation for cold start user, which gave 10% uplift Average revenue per user for cold start user.

Data Scientist

July 2022 — Nov 2022

Ads And Promotion Platform, Gojek

Bangalore

- Implemented a central store to promote re-usability of offline features across multiple use cases and act as SSOT for offline model training and online feature serving.
 - Used GitLab CI/CD to automate various functions such as BigQuery table creation, table deletion, updation of feature catalog, etc.
 - Used clockwork (Airflow Based), an internal tool, to schedule daily jobs to refresh features.
- Proposed and Designed ML architecture to introduce optimisation of bookings, in addition to clicks, in text search ads ranking. This will help in achieving high return on ad spend for ad merchants and thus resulting in higher retention.
- Designed and analysed multiple A/B experiments. This includes aligning on appropriate primary and secondary metrics, sample size calculation to ensure statistically significant results and hypothesis testing on observed data.
 - one experiment on text search ads resulted in improving ad revenue per session by 20% and was statistically significant.

Data Scientist

July 2018 — June 2022

Infrd.ai

Bangalore

- Worked on intelligent Data Capture team, Created ML architecture for training and testing loop for document information extraction from scratch to augment existing rule-based system.
- Worked on ML models to augment existing rule-based system.
- Owned End to End ML module from proof of concept to commercialization (Production).
- Applied statistical and algebraic techniques to interpret key points from gathered data.
- Developed data-gathering and reporting structures from the ground up and strategized methods capitalizing on system features.
- Used statistical technique to analyze and quantify data annotation errors.
- Object Detection on document images using CNN based models.
- Document classifier based on text and image features.

Data Scientist

Mar 2018 — Apr 2018

NanoPrecise Sci Corp

Bangalore

- Condition monitoring of Plant Asset.
- Created dashboard and deployed machine learning model using Shiny R server, AWS EC2 Instance.
- Remaining useful life calculation based on 3 axis vibration acceleration data.

Data Scientist

Factory and Fleet Analytic Team, Tata Group Technology Innovation Office

May 2017 — Feb 2018

Bangalore

- Identifying the correct data required for solving a business problem.
- Cleaning and pre-processing the dataset to the appropriate format for predictive models.
- Exploratory data analysis, to figure out important features for machine learning.
- Creating appropriate training, test, and validation set for training the model.
- Understanding the issues of underfitting, overfitting, out of sample errors.
- Flisom's (startup) Thin Film manufacturing process problem: to figure out the cause of inconsistency in module efficiencies by using a machine learning technique on process data.
- Boiler tube leak detection using principal component analysis and exponentially weighted moving average. Coal Blend (nonlinear mixture) property prediction using Machine Learning Techniques.

Research Asistant

Indian Institute of Science

2015 — 2017

Bangalore

- Solar concentrator and receiver optimisation (Monte Carlo Simulation) for supercritical CO₂ loop in the SERIUS project.

Research Asistant

Masdar Institute of Science and Technology

2013 — 2015

Abu Dhabi

- Optical modelling of a 100 kWth beam-down pilot plant for MI-Massachusetts Institute of Technology (MIT) flagship project CSPonD.
- Used Matlab to create scheme macro in lisp for TracePro, which create Heliostat and Central Reflector system geometry, translate and rotate to relevant positions, and perform Monte Carlo Ray Tracing.
- Validation of an Optical Model Applied to the Beam Down, using a heat flux sensor and charge-coupled device (CCD) camera (image processing).

Research Engineer

Alstom Hydro R&D

2010 — 2013

Vadodara

- Failure Analysis of Pelton turbine and turbine component.
- Combine hydraulic and mechanical analysis for Pelton bucket design optimization.
- Mapping of CFD output pressure to Finite Element Analysis (FEA) model at different time steps.
- Simplified the process of standard data transfer procedure between CFD and FEA.
- Performed analytical calculation for Pelton Scale Model Test Laboratory and Silt Abrasion Laboratory.
- Supported in telemetry torque measurement of Pelton scale model for calibration.
- Evaluated strain level for given loading condition through finite element calculation.
- Conducted engineering and detailed experimental tests to collect design data and assist in research work.
- Generated engineering and technical documentation to facilitate product development collaboration.

PATENT

"System To Extract Information From Documents." *"Vikas Kumar"* [link](#) (Application No: 20200184267 16/794266)
2019

PUBLICATIONS

- "Preliminary Optical, Thermal and Structural Design of a 100 kWth CSPonD Beam-down On-Sun Demonstration Plant."** *Benjamin Grange, Vikas Kumar, Peter R. Armstrong, and Daniel S. Codd* [link](#) International Conference on Applied Energy ICAE 2015 (Energy Procedia, Abu Dhabi, UAE, 2015).
2015
- "Validation of an optical model applied to the beam down CSP facility at the Masdar Institute Solar Platform"** *"Benjamin Grange, Vikas Kumar, Peter R. Armstrong, Alexander Slocum and Nicolas Calvet"* [link](#) SolarPACES 2015, Oct. 13-16, Cape Town, South Africa.
2015
- "Optical model of a complex beam down concentrator system using TracePro™"** *"Benjamin Grange, Vikas Kumar, Peter R. Armstrong, and Nicolas Calvet"* SolarPACES 2015, Oct. 13-16, Cape Town, South Africa.
2015
- "Master's Thesis: Modeling of Beam Down Solar Concentrator and Final Optical Element Design"** *"Vikas Kumar"* [link](#)
2015

PROJECT SUMMARY

Information Extraction	Python	Information extraction from Document using deep learning techniques.
Scoring System	Python	Scoring system for information extraction using machine learning
Document Classification	Python	Document classification for legal documents using deep learning
Scoring System	Python	Scoring system for tenant and landlord onboarding for Zibo

EDUCATION

Master of Science , <i>Masdar Institute of Science and Technology</i>	2015
Btech , <i>NIT Kurukshetra</i>	2009

CERTIFICATIONS

Machine Learning, Coursera	2013
Deep Learning, deeplearning.ai	2017
Certificate in Engineering Excellence in Big Data Analytics and Optimization, Insofe	2016
Machine Learning, ISI Bangalore	2020